

Rules for computing derivatives



Outline

Algebra of derivatives

Derivative of a composite function

Derivative of the inverse function

Logarithmic derivative and elasticity

Algebra of derivatives



Introduction: Algebra of derivatives

- ▶ We now look at the relation between the derivative operation and the main operations on functions that we already know; in particular we will show the relation between:
 1. differentiation and algebraic operations
 2. differentiation and composition
 3. differentiation and inversion
- ▶ In this section we start from the **algebraic operations**: sum, difference, product and quotient of differentiable functions.
- ▶ These rules allow us to differentiate a great many functions without resorting to the definition each time.

Theorem on the algebra of derivatives

Theorem 1: algebra of derivatives

Let $f, g: (a, b) \rightarrow \mathbb{R}$ be two functions differentiable in (a, b) .

Then $f \pm g$, $f \cdot g$, f/g ($g \neq 0$) are differentiable in (a, b) and the following formulas hold:

$$(f \pm g)' = f' \pm g' \quad (\text{D1})$$

$$(f \cdot g)' = f' \cdot g + f \cdot g' \quad (\text{D2})$$

$$\left(\frac{f}{g}\right)' = \frac{f' \cdot g - f \cdot g'}{g^2} \quad (g \neq 0) \quad (\text{D3})$$

Consequences of the rules

- From rule (D2) we deduce

$$(k \cdot g)' = k \cdot g' \quad \text{with } k \in \mathbb{R} \text{ constant} \quad (\text{D4})$$

since the derivative of a constant is equal to zero.

- From rule (D3) we deduce

$$\left(\frac{1}{g}\right)' = -\frac{g'}{g^2} \quad (g \neq 0) \quad (\text{D5})$$

Leibniz rule

- ▶ Rule (D2) is called the **Leibniz rule** (product rule) and it extends to the product of n functions:

$$(f_1 f_2 \cdots f_n)' = f_1' f_2 \cdots f_n + f_1 f_2' \cdots f_n + \cdots + f_1 f_2 \cdots f_n'$$

- ▶ The theorem actually has a pointwise character: that is, if f and g are differentiable at a point $x_0 \in (a, b)$, then at that point $f \pm g$, $f \cdot g$, f/g are also differentiable, and the formulas above hold (and they extend to intervals).

Proof of rule (D2) (1/2)

We prove rule (D2). Fixing $x \in (a, b)$, we have

$$\begin{aligned} f(x+h)g(x+h) - f(x)g(x) &= \\ &= f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x), \end{aligned}$$

and hence the difference quotient can be written as:

$$\begin{aligned} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} &= \\ &= f(x+h) \underbrace{\frac{g(x+h) - g(x)}{h}}_{\rightarrow g'(x)} \\ &\quad + g(x) \underbrace{\frac{f(x+h) - f(x)}{h}}_{\rightarrow f'(x)} \end{aligned}$$

Proof of rule (D2) (2/2)

We have

$$f(x+h) \rightarrow f(x) \text{ for } h \rightarrow 0,$$

since f is continuous, being differentiable.

Hence

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h} \rightarrow f(x)g'(x) + f'(x)g(x) \text{ as } h \rightarrow 0.$$



Proof of rule (D5)

We prove rule (D5). Fixing $x \in (a, b)$, the difference quotient can be written as:

$$\begin{aligned}\frac{1}{h} \left[\frac{1}{g(x+h)} - \frac{1}{g(x)} \right] &= \frac{g(x) - g(x+h)}{h g(x) g(x+h)} \\ &= - \frac{g(x+h) - g(x)}{h} \cdot \frac{1}{g(x) g(x+h)}\end{aligned}$$

We have

$$g(x+h) \rightarrow g(x) \quad \text{for } h \rightarrow 0,$$

since g is continuous, being differentiable. Hence

$$\frac{1}{h} \left[\frac{1}{g(x+h)} - \frac{1}{g(x)} \right] \rightarrow - \frac{g'(x)}{g^2(x)} \quad \text{for } h \rightarrow 0.$$



Proof of rule (D3)

Note that rule (D3) follows from rules (D2) and (D5); indeed:

$$\begin{aligned}\left(\frac{f(x)}{g(x)}\right)' &= \left(f(x) \cdot \frac{1}{g(x)}\right)' = \underbrace{f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \left(\frac{1}{g(x)}\right)'}_{\text{rule (D2)}} \\ &= f'(x) \cdot \frac{1}{g(x)} + f(x) \cdot \underbrace{\left(-\frac{g'(x)}{g^2(x)}\right)}_{\text{rule (D5)}} \\ &= \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g^2(x)}.\end{aligned}$$



Derivative of the tangent

Observation 1

Given the function $f(x) = \tan x$, the derivative function is $f'(x) = \frac{1}{\cos^2 x} = 1 + \tan^2 x$.

Proof

Using formula (D3) we have

$$f'(x) = \left(\frac{\sin x}{\cos x} \right)' = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = 1 + \tan^2 x$$



Derivative of the cotangent

Observation 2

Given the function $f(x) = \cot x$, the derivative function is $f'(x) = -\frac{1}{\sin^2 x} = -(1 + \cot^2 x)$.

Proof

Using formula (D3) we have

$$\begin{aligned}f'(x) &= \left(\frac{\cos x}{\sin x}\right)' = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\frac{\sin^2 x + \cos^2 x}{\sin^2 x} \\&= -\frac{1}{(\sin^2 x)} = -(1 + \cot^2 x)\end{aligned}$$



Summary

Algebra of derivatives:

- ▶ $(f \pm g)' = f' \pm g', \quad (k g)' = k g'$
- ▶ $(f \cdot g)' = f' \cdot g + f \cdot g'$ (Leibniz rule)
- ▶ $\left(\frac{f}{g}\right)' = \frac{f' \cdot g - f \cdot g'}{g^2}, \quad \left(\frac{1}{g}\right)' = -\frac{g'}{g^2}$
- ▶ $(\tan x)' = \frac{1}{\cos^2 x} = 1 + \tan^2 x, \quad (\cot x)' = -\frac{1}{\sin^2 x}$

Derivative of a composite function



Introduction: Derivative of a composite function

- ▶ In this section we study the derivative of the **composition** of two functions: the **chain rule**.
- ▶ It is one of the most used rules: it allows us to differentiate functions such as $\sin^3 x$, $e^{-\alpha x} \cos(\omega x + \varphi)$, a^x , $f(x)^{g(x)}$, $|f(x)|$.
- ▶ We will also see its form in Leibniz's notation and many examples.

Theorem 2: chain rule

Let $g \circ f$ be the composite function of two functions f and g . If f is differentiable at a point x and g is differentiable at $y = f(x)$, then $g \circ f$ is differentiable at x and the following formula holds:

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) \quad (\text{C1})$$

Proof (1/3)

We have

$$(g \circ f)(x + h) - (g \circ f)(x) = g(f(x + h)) - g(f(x))$$

If we set

$$k = f(x + h) - f(x) \quad \text{and} \quad y = f(x) \quad \text{then} \quad f(x + h) = y + k$$

By the continuity of f , $h \rightarrow 0$ implies $k \rightarrow 0$. With the new notation we have:

$$g(f(x + h)) - g(f(x)) = g(y + k) - g(y)$$

We now observe that the definition of derivative:

$$g'(y) = \lim_{k \rightarrow 0} \frac{g(y + k) - g(y)}{k}$$

Proof (2/3)

can be rewritten, for $k \neq 0$, as

$$\frac{g(y+k) - g(y)}{k} = g'(y) + \varepsilon(k)$$

where $\varepsilon(k)$ denotes a quantity that tends to zero as $k \rightarrow 0$.

Multiplying both sides of the previous equation by k we find

$$g(y+k) - g(y) = k \cdot g'(y) + k \cdot \varepsilon(k)$$

Proof (3/3)

This relation also holds for $k = 0$. Therefore:

$$g(f(x+h)) - g(f(x)) = k \cdot g'(y) + k \cdot \varepsilon(k)$$

Dividing by h , and observing that

$$\frac{k}{h} \rightarrow f'(x) \quad \text{for} \quad h \rightarrow 0$$

which implies, as seen above, also $k \rightarrow 0$, we obtain the claim. $\square$

Composition of several functions

- ▶ As the name “chain rule” suggests, (C1) can be generalized to the composition of any number of functions, composed one with another.
- ▶ For example, for three functions we have:

$$\left(f\left(g\left(h(x) \right) \right) \right)' = f'\left(g\left(h(x) \right) \right) \cdot g'\left(h(x) \right) \cdot h'(x)$$

Example: derivative of a composite function

We compute the derivative function of the composite function:

$$h(x) = \underbrace{\sin^3 x}_{=(g \circ f)(x)}$$

The two functions are:

$$f(x) = \sin x \quad \text{and} \quad g(y) = y^3 \quad \text{with} \quad y = \sin x.$$

Both functions are differentiable on all of $\mathbb{R}$. We have

$$g'(y) = 3y^2 \quad \text{and} \quad f'(x) = \cos x$$

hence, using the chain rule (C1) and substituting back, we obtain:

$$h'(x) = 3 \sin^2 x \cdot \cos x.$$

Chain rule in Leibniz's notation

Consider the composite function $g(f(x))$, equivalently written $(g \circ f)(x)$. Setting $y = f(x)$ and $w = g(y)$, and using the (Leibniz) notation

$\frac{df}{dx}$ and $\frac{dg}{dy}$ for the derivatives of f and g w.r.t. x

the (chain) rule (C1) can be rewritten as:

$$\frac{dw}{dx} = \frac{dw}{dy} \cdot \frac{dy}{dx} \quad (\text{C2})$$

Rule (C2) expresses the fact that the rate of change of w with respect to x is the product of the “intermediate” rates of change, of w with respect to y and of y with respect to x .

Example: derivative of a composite function

Consider the composite function of the previous exercise. Setting

$$y = \sin x \quad (y = f(x)) \quad \text{and} \quad w = y^3 \quad (w = g(y))$$

and using Leibniz's notation, rule (C2) would be written:

$$\frac{dw}{dx} = \frac{dw}{dy} \cdot \frac{dy}{dx} = 3y^2 \cdot \cos x.$$

And hence, substituting $y = \sin x$, we obtain:

$$\frac{dw}{dx} = 3 \sin^2 x \cdot \cos x.$$

Example: derivative of a composite function

We compute the derivative function of the composite function (multiplied by a constant):

$$h(x) = A \cdot \underbrace{\sin(\omega x + \varphi)}_{=(g \circ f)(x)} \quad \text{with} \quad A, \omega, \varphi \in \mathbb{R}.$$

The two functions are:

$$f(x) = \omega x + \varphi \quad \text{and} \quad g(y) = \sin y \quad \text{with} \quad y = \omega x + \varphi.$$

Both functions are differentiable on all of $\mathbb{R}$. We have:

$$f'(x) = \omega \quad \text{and} \quad g'(y) = \cos y$$

hence, using rule (C1), rule (D4) and substituting back, we obtain:

$$h'(x) = A \cdot \left(\cos(\omega x + \varphi) \cdot \omega \right)$$

Example: product of composite functions (1/2)

We compute the derivative function of the product of two composite functions times a constant:

$$h(x) = A \cdot \underbrace{e^{-\alpha x}}_{=(r \circ s)(x)} \cdot \underbrace{\cos(\omega x + \varphi)}_{=(g \circ f)(x)} \quad \text{with} \quad A, \omega, \varphi \in \mathbb{R}, \alpha \in \mathbb{R}_+.$$

The functions are:

$$\begin{aligned} s(x) &= -\alpha x & \text{and} & & r(z) &= e^z & \text{with} & & z &= -\alpha x; \\ f(x) &= \omega x + \varphi & \text{and} & & g(y) &= \cos y & \text{with} & & y &= \omega x + \varphi. \end{aligned}$$

The functions are differentiable on all of $\mathbb{R}$. We have:

$$s'(x) = -\alpha \quad \text{and} \quad r'(z) = e^z; \quad f'(x) = \omega \quad \text{and} \quad g'(y) = -\sin y$$

Example: product of composite functions (2/2)

Consequently, using the chain rule (C1) and substituting back, we obtain:

$$(r \circ s)'(x) = -\alpha e^{-\alpha x}; \quad (g \circ f)'(x) = -\sin(\omega x + \varphi) \cdot \omega$$

Hence, using rule (D2) and rule (D4):

$$\begin{aligned} h'(x) = A \cdot & \left(\underbrace{-\alpha e^{-\alpha x}}_{(r \circ s)'(x)} \cdot \underbrace{\cos(\omega x + \varphi)}_{(g \circ f)(x)} \right. \\ & \left. + \underbrace{e^{-\alpha x}}_{(r \circ s)(x)} \cdot \underbrace{\left(-\sin(\omega x + \varphi) \cdot \omega \right)}_{(g \circ f)'(x)} \right) \end{aligned}$$

Example: derivative of a composite function

Let $f(x) > 0$ be differentiable; we compute the derivative of the function

$$h(x) = \log(f(x))$$

using the chain rule (C1) we have:

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$

Derivative of the exponential function with base a

Observation 3

Given the function $f(x) = a^x$, with $a > 0$, the derivative function is $f'(x) = a^x \cdot \log a$.

Proof

We have

$$f'(x) = \left(\exp(x \cdot \log a) \right)'$$

hence, using the formulas for the derivative of e^y (with $y = x \log a$) and for the derivative of a composite function, we have

$$f'(x) = \exp(x \cdot \log a) \cdot \log a = a^x \cdot \log a$$



Derivative of the logarithm with base a

Observation 4

Given the function $f(x) = \log_a x$, with $a > 0, a \neq 1$, the derivative function is $f'(x) = \frac{1}{x \log a}$.

Proof

We have

$$f'(x) = \left(\frac{\log x}{\log a} \right)'$$

hence, using the formulas for the derivative of $\log x$ and for the derivative of $k f(x)$, we have:

$$f'(x) = \frac{1}{\log a} \cdot \frac{1}{x} = \frac{1}{x \log a}$$



Derivative of the hyperbolic sine

Observation 5

Given the function $f(x) = \sinh x$, the derivative function is $f'(x) = \cosh x$.

Proof

We have:

$$f'(x) = \left(\frac{e^x - e^{-x}}{2} \right)' = \frac{1}{2} \left(e^x - (-e^{-x}) \right) = \frac{e^x + e^{-x}}{2} = \cosh x$$



Derivative of the hyperbolic cosine

Observation 6

Given the function $f(x) = \cosh x$, the derivative function is $f'(x) = \sinh x$.

Proof

We have:

$$f'(x) = \left(\frac{e^x + e^{-x}}{2} \right)' = \frac{1}{2} \left(e^x + (-e^{-x}) \right) = \frac{e^x - e^{-x}}{2} = \sinh x$$



Derivative of $f(x)^{g(x)}$

The derivatives of functions of the type:

$$h(x) = f(x)^{g(x)}$$

are based on rewriting the function (with $f(x) > 0$) in the following form:

$$f(x)^{g(x)} = \exp \left(g(x) \cdot \log (f(x)) \right)$$

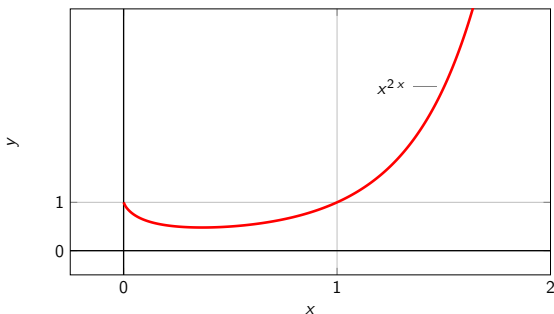
We have:

$$\begin{aligned} \left(f(x)^{g(x)} \right)' &= \left(\exp (g(x) \cdot \log (f(x))) \right)' \\ &= \exp (g(x) \cdot \log (f(x))) \cdot \left(g(x) \cdot \log (f(x)) \right)' \\ &= f(x)^{g(x)} \cdot \left(g'(x) \cdot \log (f(x)) + g(x) \cdot \frac{f'(x)}{f(x)} \right) \end{aligned}$$

Example: function raised to a function (1/2)

We compute the derivative of the function

$$h(x) = x^{2x} = \exp(2x \cdot \log x) \quad \text{with } x > 0$$



Example: function raised to a function (2/2)

The two functions are:

$$f(x) = x \quad \text{and} \quad g(x) = 2x$$

Hence:

$$\begin{aligned} h'(x) &= \left(\exp(2x \cdot \log x) \right)' = \exp(2x \cdot \log x) \cdot (2x \cdot \log x)' \\ &= x^{2x} \cdot \left(2 \log x + 2x \frac{1}{x} \right) = x^{2x} (2 \log x + 2) \end{aligned}$$

Derivative of the absolute value of a function

Consider the absolute value of a function: $|f(x)|$.

At the points where $f(x) \neq 0$, the differentiation of a composite function gives:

$$(|f(x)|)' = \operatorname{sgn}(f(x)) \cdot f'(x) = \begin{cases} f'(x) & \text{for } f(x) > 0 \\ -f'(x) & \text{for } f(x) < 0 \end{cases}$$

In general, we expect the function $|f(x)|$ to have corner points at the points where $f(x)$ vanishes.

Example: derivative with absolute value (1/3)

We compute the derivative function of the function:

$$f(x) = |x^2 - 4x + 3|, \quad \text{with } x \neq 1 \text{ and } x \neq 3$$

we have

$$f'(x) = \operatorname{sgn}(x^2 - 4x + 3) \cdot (2x - 4)$$

equivalently:

$$f'(x) = \begin{cases} 2x - 4, & x < 1 \text{ or } x > 3 & \text{i.e. for } f(x) > 0 \\ -(2x - 4) & 1 < x < 3 & \text{i.e. for } f(x) < 0 \end{cases}$$

Example: derivative with absolute value (2/3)

At $x_0 = 1$, we need to use the limit of the difference quotient:

$$\begin{aligned}\frac{f(x_0 + h) - f(x_0)}{h} &= \frac{f(1 + h) - f(1)}{h} \\ &= \frac{|(1 + h)^2 - 4(1 + h) + 3|}{h} = \frac{|h^2 - 2h|}{h}\end{aligned}$$

If $h \rightarrow 0^-$ we have $|h^2 - 2h| = h^2 - 2h$, hence:

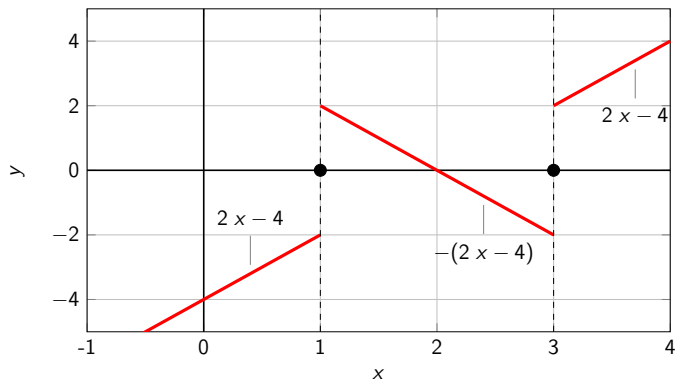
$$\lim_{h \rightarrow 0^-} \frac{h^2 - 2h}{h} = -2$$

If $h \rightarrow 0^+$ we have $|h^2 - 2h| = -h^2 + 2h$, hence:

$$\lim_{h \rightarrow 0^+} \frac{-h^2 + 2h}{h} = 2$$

The function is not differentiable at $x_0 = 1$ and $x_0 = 3$ (by arguments analogous to those for $x_0 = 1$), where it has corner points.

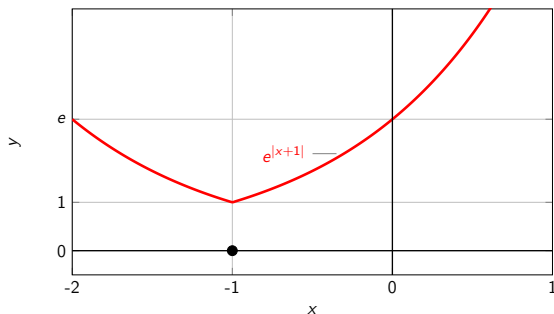
Example: derivative with absolute value (3/3)



Example: derivative with absolute value (1/3)

We compute the derivative of the function:

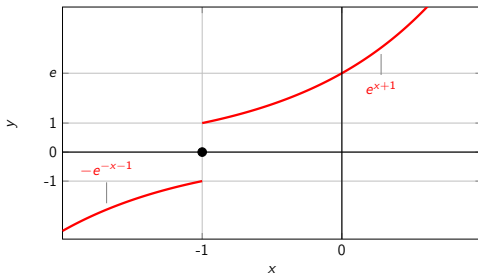
$$f(x) = e^{|x+1|}$$



Example: derivative with absolute value (2/3)

For $x \neq -1$, we have:

$$f(x) = \begin{cases} e^{x+1}, & x > -1 \\ e^{-x-1}, & x < -1 \end{cases} \quad \text{hence} \quad f'(x) = \begin{cases} e^{x+1}, & x > -1 \\ -e^{-x-1}, & x < -1 \end{cases}$$



Example: derivative with absolute value (3/3)

At $x_0 = -1$, we need to use the limit of the difference quotient:

$$\begin{aligned}\frac{f(x_0 + h) - f(x_0)}{h} &= \frac{f(-1 + h) - f(-1)}{h} \\ &= \frac{e^{|-1+h+1|} - 1}{h} = \frac{e^{|h|} - 1}{h}\end{aligned}$$

We have:

$$\text{if } h \rightarrow 0^+, e^{|h|} = e^h \text{ and } \lim_{h \rightarrow 0^+} \frac{e^h - 1}{h} = 1$$

$$\text{if } h \rightarrow 0^-, e^{|h|} = e^{-h} \text{ and } \lim_{h \rightarrow 0^-} \frac{e^{-h} - 1}{h} = -1$$

We conclude that, since the limit of the difference quotient does not exist, the function is not differentiable at $x_0 = -1$. The function has a corner point at $x_0 = -1$.

Derivatives of some logarithmic functions (1/2)

We now consider the derivatives of some logarithmic functions. We compute:

$$\left(\log(|x|) \right)' = \frac{1}{|x|} \cdot \operatorname{sgn}(x) = \frac{1}{x}$$

Moreover, by the differentiation of composite functions we have:

$$\left(\log(|f(x)|) \right)' = \frac{1}{|f(x)|} \cdot \operatorname{sgn}(f(x)) \cdot f'(x) = \frac{f'(x)}{f(x)}$$

Derivatives of some logarithmic functions (2/2)

Sometimes it is convenient to use the properties of logarithms to transform a logarithmic function before computing its derivative function:

$$(\log(cx))' = (\log(c) + \log(x))' = \frac{1}{x} \quad \text{with } c \in \mathbb{R}$$

$$\begin{aligned} \left(\log \left(\frac{ax+b}{cx+d} \right) \right)' &= (\log(ax+b) - \log(cx+d))' \\ &= \frac{a}{ax+b} - \frac{c}{cx+d} \quad \text{with } a, b, c, d \in \mathbb{R} \end{aligned}$$

Summary

Derivative of a composite function:

- ▶ Chain rule: $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$; with Leibniz
$$\frac{dw}{dx} = \frac{dw}{dy} \cdot \frac{dy}{dx}$$
- ▶ $(a^x)' = a^x \log a$, $(\log_a x)' = \frac{1}{x \log a}$, $(\sinh x)' = \cosh x$,
 $(\cosh x)' = \sinh x$
- ▶ $f(x)^{g(x)} = \exp(g(x) \log f(x))$
- ▶ $(|f(x)|)' = \operatorname{sgn}(f(x)) \cdot f'(x)$ where $f(x) \neq 0$; possible corner points where $f(x) = 0$
- ▶ $(\log |f(x)|)' = \frac{f'(x)}{f(x)}$

Derivative of the inverse function



Introduction: Derivative of the inverse function

- ▶ In this section we study the relation between the derivative of an invertible function and that of its **inverse function**.
- ▶ We will see the geometric meaning of the formula, related to the symmetry of the graphs with respect to the bisector $y = x$.
- ▶ We will derive the derivatives of $\arctan$, $\arcsin$, $\arccos$, SettSh , SettCh , and we will see how to compute the derivative of the inverse even when we cannot write it explicitly.

Theorem on the derivative of the inverse function

Theorem 3: derivative of the inverse function

Let $f : (a, b) \rightarrow \mathbb{R}$ be a continuous and invertible function in (a, b) and $g = f^{-1}$ its inverse function, defined in $f((a, b))$.

Assume moreover that $f'(x_0) \neq 0$ exists for some $x_0 \in (a, b)$.

Then g is differentiable at $y_0 = f(x_0)$ and

$$g'(y_0) = \frac{1}{f'(x_0)} \quad (I)$$

Proof (1/2)

Let:

$$f(x_0) = y_0; \quad g(y_0) = x_0$$

$$f(x_0 + h) = y_0 + k; \quad g(y_0 + k) = x_0 + h$$

Consider the difference quotient of g at y_0 :

$$\frac{g(y_0 + k) - g(y_0)}{k} = \frac{h}{f(x_0 + h) - f(x_0)}$$

If $k \neq 0$, then $f(x_0 + h) - f(x_0) \neq 0$ and hence also $h \neq 0$; therefore the last quotient can also be rewritten in the form:

$$\frac{1}{\frac{f(x_0 + h) - f(x_0)}{h}}$$

Proof (2/2)

Moreover, for $k \rightarrow 0$ we have $g(y_0 + k) \rightarrow g(y_0)$ because g is continuous, being the inverse of a continuous function on an interval (theorem on the continuity of inverse functions); on the other hand $h = g(y_0 + k) - g(y_0)$, hence for $k \rightarrow 0$ also $h \rightarrow 0$, and by hypothesis

$$\frac{1}{\frac{f(x_0+h)-f(x_0)}{h}} \rightarrow \frac{1}{f'(x_0)} \quad \text{for } h \rightarrow 0$$

hence the following limit exists

$$\lim_{k \rightarrow 0} \frac{g(y_0 + k) - g(y_0)}{k} = \frac{1}{f'(x_0)}$$



A remark on the proof

- ▶ We observe that, assuming the differentiability of f^{-1} , (I) follows immediately from the identity

$$g(f(x)) = x$$

and from the chain rule:

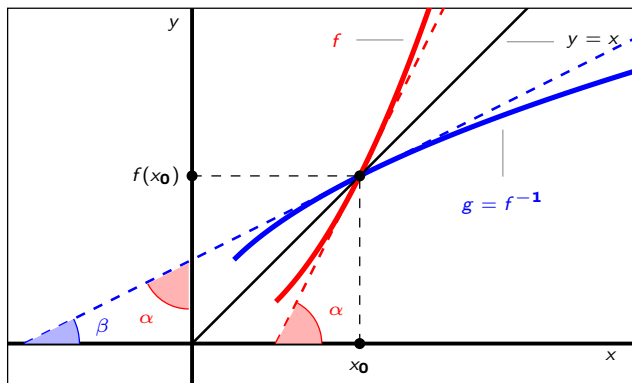
$$g'(f(x)) \cdot f'(x) = 1$$

from which, if $f'(x) \neq 0$, (I) follows.

- ▶ The proof we have given, however, is necessary in order to deduce the differentiability of g from our hypotheses on f .

Geometric meaning (1/2)

(I) has a simple geometric meaning, recalling that the graphs of f and $g = f^{-1}$ are symmetric with respect to the bisector $y = x$. Geometrically we have:



Geometric meaning (2/2)

Hence the angles α and β are complementary, that is:

$$\left(\alpha + \beta = \frac{\pi}{2}\right)$$

and hence

$$f'(x) = \tan \alpha = \tan \left(\frac{\pi}{2} - \beta\right) = \frac{1}{\tan \beta} = \frac{1}{g'(y)}$$

With Leibniz's notation, setting $y = f(x)$ and $x = g(y)$, (I) is written in the form

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

- Pay attention to the fact that in the formula for the derivative of the inverse function, the derivatives f' and g' are computed at two different points: this is the main thing to be careful about when applying this theorem.

Derivative of the arctangent

Observation 7

Given the function $f(y) = \arctan y$, the derivative function is $f'(y) = \frac{1}{1+y^2}$.

Proof

We set $y = \tan x$, $x = \arctan y$ with $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $y \in \mathbb{R}$. We have:

$$(\arctan y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{1 + \tan^2 x} = \frac{1}{1 + y^2}$$



Derivative of the arcsine

Observation 8

Given the function $f(y) = \arcsin y$, the derivative function is
$$f'(y) = \frac{1}{\sqrt{1-y^2}}$$

Proof

We set $y = \sin x$, $x = \arcsin y$, with $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $y \in [-1, 1]$.
Since for those values of x we have

$$\cos x = \sqrt{1 - \sin^2 x} = \sqrt{1 - y^2}$$

we have:

$$(\arcsin y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\cos x} = \frac{1}{\sqrt{1 - y^2}}$$



Derivative of the arccosine

Observation 9

Given the function $f(y) = \arccos y$, the derivative function is
$$f'(y) = -\frac{1}{\sqrt{1-y^2}}$$

Proof

Now we set $y = \cos x$, $x = \arccos y$, with $x \in [0, \pi]$, $y \in [-1, 1]$.
Since for those values of x we have

$$\sin x = \sqrt{1 - \cos^2 x} = \sqrt{1 - y^2}$$

we have

$$(\arccos y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{-\sin x} = -\frac{1}{\sqrt{1 - y^2}}$$



Usefulness of the theorem

- Note that the functions $\arcsin x$, $\arccos x$, although defined and continuous on $[-1, 1]$, are not differentiable at the endpoints of the interval: precisely, at these points they have a vertical tangent.

The usefulness of the theorem on the derivative of the inverse function lies in the fact that it allows us to compute the derivative of the inverse function g even in situations where we do not know how to write g explicitly.

Example: derivative of the inverse function

Let

$$f(x) = x + e^x$$

The function is strictly increasing on all of $\mathbb{R}$, hence invertible; let g be its inverse. We compute, for example, $g'(y_0)$ for $y_0 = f(0) = 1$. We have:

$$f'(x) = 1 + e^x, \quad f'(0) = 2 \neq 0$$

Hence:

$$g'(1) = \frac{1}{f'(0)} = \frac{1}{2}.$$

Derivative of the inverse hyperbolic sine

Observation 10

Given the function $f(y) = \text{SettSh } y$, the derivative function is

$$f'(y) = \frac{1}{\sqrt{y^2+1}}$$

Proof

Now we set $y = \operatorname{Sh} x$, $x = \operatorname{SettSh} y$, with $x \in \mathbb{R}$, $y \in \mathbb{R}$. Since for those values of x we have

$$\operatorname{Ch} x = \sqrt{\operatorname{Sh}^2 x + 1} = \sqrt{y^2 + 1}$$

we have:

$$(\operatorname{SettSh} y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\operatorname{Ch} x} = \frac{1}{\sqrt{y^2 + 1}}$$



Derivative of the inverse hyperbolic cosine

Observation 11

Given the function $f(y) = \text{SettCh } y$, the derivative function is

$$f'(y) = \frac{1}{\sqrt{y^2-1}}$$

Proof

Now we set $y = \operatorname{Ch} x$, $x = \operatorname{SettCh} y$, with $x \in [0, +\infty)$, $y \in [1, +\infty)$. Since for those values of x we have

$$\operatorname{Sh} x = \sqrt{\operatorname{Ch}^2 x - 1} = \sqrt{y^2 - 1}$$

we have:

$$(\operatorname{SettCh} y)' = \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \frac{1}{\operatorname{Sh} x} = \frac{1}{\sqrt{y^2 - 1}}$$



Summary

Derivative of the inverse function:

- ▶ $g = f^{-1}$, $f'(x_0) \neq 0$: $g'(y_0) = \frac{1}{f'(x_0)}$ with $y_0 = f(x_0)$; with Leibniz $\frac{dx}{dy} = \frac{1}{dy/dx}$
- ▶ The derivatives of f and g are computed at **different points**
- ▶ $(\arctan y)' = \frac{1}{1+y^2}$, $(\arcsin y)' = \frac{1}{\sqrt{1-y^2}}$,
 $(\arccos y)' = -\frac{1}{\sqrt{1-y^2}}$
- ▶ $(\text{SettSh } y)' = \frac{1}{\sqrt{y^2+1}}$, $(\text{SettCh } y)' = \frac{1}{\sqrt{y^2-1}}$

Logarithmic derivative and elasticity



Introduction: Logarithmic derivative and elasticity

- ▶ In this section we introduce the **logarithmic derivative**, which measures the **relative** growth rate of a function.
- ▶ We will see graphs on semi-logarithmic and logarithmic scales.
- ▶ Finally we will define **elasticity**, widely used for example in economics.

Logarithmic derivative

Definition 1: of logarithmic derivative

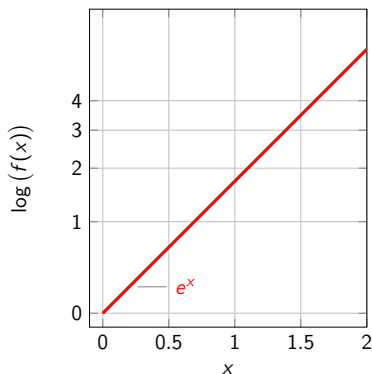
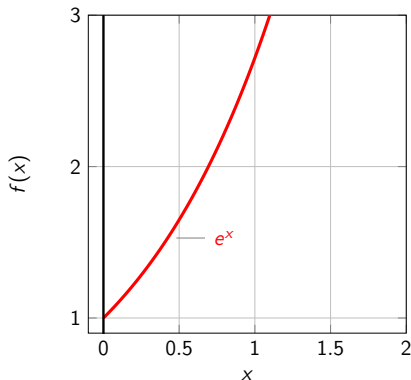
Given $f > 0$, the logarithmic derivative function of f is the derivative function of $\log f$; in formulas:

$$\frac{d}{dx} \log (f(x)) = \frac{f'(x)}{f(x)}$$

- ▶ The logarithmic derivative has the meaning of **relative growth rate** of f with respect to x .
- ▶ In some cases the relative growth rate $\frac{f'(x)}{f(x)}$ is often more meaningful than the absolute rate $f'(x)$.
- ▶ For example, a yearly capital increase of 1 billion out of a total of 10 has a very different effect from an increase of 1 billion out of 1000 billion. In the first case the relative rate of change is $1/10$, in the second $1/1000$, while the absolute increases are equal.

Semi-logarithmic scale

When we are interested in visualizing relative increases, we use graphs on a **semi-logarithmic scale**: the values of x are placed on the x -axis, while the values of $\log(f(x))$ are placed on the y -axis.



Logarithmic scale

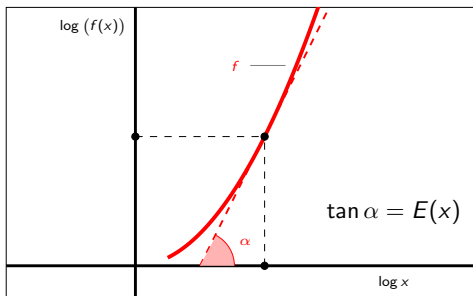
- ▶ This kind of representation is also used when $f(x)$ grows so rapidly that it would require an excessive compression of the scale (unit of measure) on the y -axis (or on the x -axis).

A **log-log graph** (graph on a logarithmic scale) is a graph in which, instead of x , the values of $\log x$ are placed on the x -axis and, instead of $f(x)$, the values of $\log f(x)$ are placed on the y -axis.

Definition 2: of elasticity

The slope of the tangent line to a graph on a logarithmic scale is called the elasticity of f and is denoted by $E(x)$.

- Elasticity represents the relative rate of change of f with respect to relative changes of x , and it equals the derivative of $\log f$ with respect to $\log x$.



Analytic expression of elasticity

- To find the analytic expression of $E(x)$, we observe that, by the theorem on the differentiation of composite functions, setting $u = \log x$ we have:

$$\frac{d \log f}{dx} = \frac{d \log f}{du} \frac{du}{dx} \quad \text{that is} \quad \frac{f'(x)}{f(x)} = E(x) \cdot \frac{1}{x}$$

from which we obtain:

$$E(x) = x \cdot \frac{f'(x)}{f(x)}$$

Example: elasticity

We compute the elasticity of the power function $f(x) = x^\alpha$, $x > 0$, $\alpha \in \mathbb{R}$; we have:

$$E(x) = x \cdot \frac{\alpha x^{\alpha-1}}{x^\alpha} = \alpha \quad (\text{constant elasticity})$$

Logarithmic derivative and elasticity:

- ▶ Logarithmic derivative: $\frac{d}{dx} \log f(x) = \frac{f'(x)}{f(x)}$, relative growth rate
- ▶ Semi-logarithmic scale: x and $\log f(x)$; logarithmic scale: $\log x$ and $\log f(x)$
- ▶ Elasticity: $E(x) = x \cdot \frac{f'(x)}{f(x)}$, slope on a logarithmic scale
- ▶ The power x^α has constant elasticity α

Key messages

Summary

1. Algebraic operations:

- ▶ Sums, products (Leibniz) and quotients of differentiable functions are differentiable

2. Composition:

- ▶ Chain rule: $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$

3. Inversion:

- ▶ $g'(y_0) = \frac{1}{f'(x_0)}$ with $y_0 = f(x_0)$ and $f'(x_0) \neq 0$

4. Logarithmic derivative and elasticity:

- ▶ $\frac{f'(x)}{f(x)}$ and $E(x) = x \cdot \frac{f'(x)}{f(x)}$